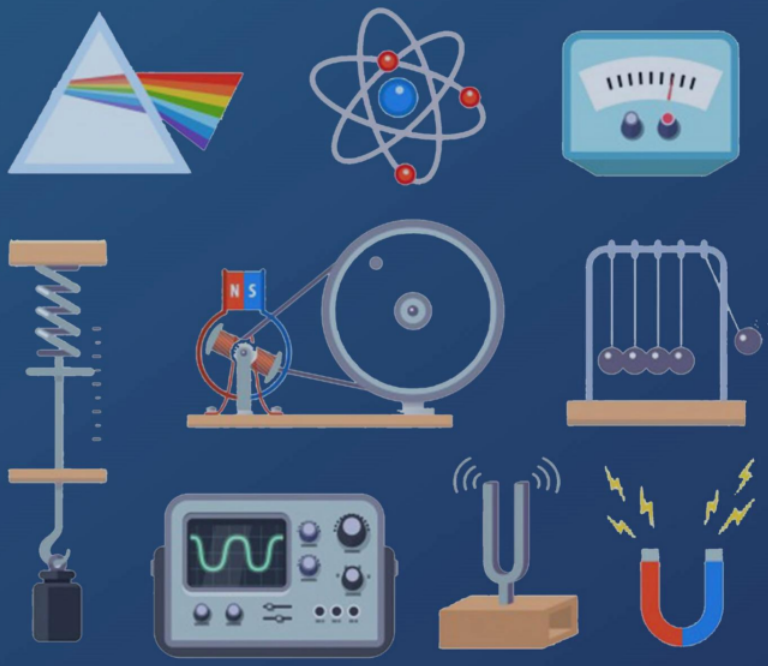
PHYSICS

Electromagnetic Induction (EMI)



SHEET

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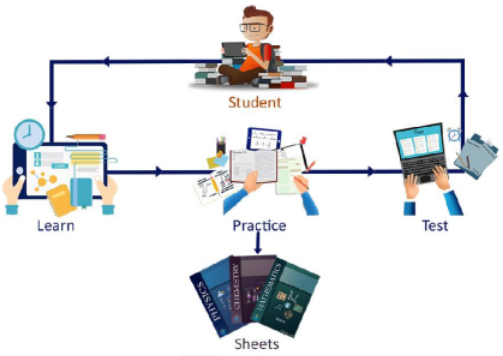
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	EMI - Introduction				
2	Magnetic Flux	1	17-19		
3	Faraday's Law of Induction	2-5	1	1	1
4	Lenz's Law	6-12	2-4,16, 20-22	2-5	8,9
5	Motional EMF	13-16	5,6, 13-15,23	6-10	2,13
6	Eddy Currents	17,18	7		
7	Induced Electric Field	19	8	11	
8	Self Inductance	20-22	9	12,13	3
9	Growth Of Current In LR Circuit	23-24		14-17	4
10	Energy Stored In Inductor	25,26	10	18	
11	Decay Of Current In LR Circuit	27		19	10,14,15
12	Mutual Inductance	28,29	11	20	6
13	Inductors In Series And Parallel	30		21	12
14	AC Generator	31,32		22	7,11,16
15	After Full Chapter		12		5

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Magnetic Flux

Q1. A square of side *2* meters lies in the *x-y* plane in a region, where the magnetic field is given by $\vec{B} = B_0\left(2\hat{i} + 3\hat{j} + 4\hat{k}\right)T$, where B_o is constant. The magnitude of flux passing through the square is -

(A) $8B_o$ Wb. (B) $12B_o$ Wb.

(C) $16B_o$ Wb. (D) $\sqrt{4\times 29}B_0 Wb$

Faraday's Law of Induction

Q2. The dimension of the ratio of magnetic flux and the resistance is equal to that of :

(A) induced emf (B) charge

(C) inductance (D) current

Q3. In a circuit a coil of resistance 2Ω, then magnetic flux changes from 2.0Wb to 10.0Wb in 0.2 sec. The charge flow in the coil during this time is :-

(A) 5.0 C (B) 4.0 C

(C) 1.0 C (D) 0.8 C

Q4. According to Faraday's Laws of electro magnetic induction :–

(A) the direction of the induced current is such that it opposes the causes of change in flux.

(B) the induced emf in the coil is equal to the rate of change of magnetic flux associated with it

(C) the direction of induced emf is such that it opposes the causes of change in flux.

(D) none of the above

Q5. A coil of 100 turns having an average area of 100 cm^2 for each turn is held in a uniform field of 50 gauss, The direction of field being at right angles to the plane of coil. The field is removed in $\frac{1}{100}$ s. Average emf (in volts) induced in the coil is :-

(A) 0.5 V (B) 10 V

(C) 15 V (D) 50 V

Lenz's Law

Q6. The flux linked with a coil at any instant 't' is given by : $\phi = 10t^2 - 50t + 250$. The induced emf at t = 3s is-

(A) – 190 V (B) – 10 V

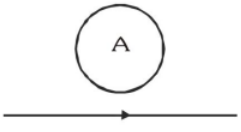
(C) 10 V (D) 190 V

Q7. A coil of resistance 1000Ω and 1000 turns have the magnetic flux line of 5.5×10^{-5} Wb. If the magnetic flux is changed to 5×10^{-5} Wb. in 0.1 sec. then the induced charge in coil is :–

(A) 50 μC (B) 5 μC

(C) 2 μC (D) 20 μC

Q8. As shown in the figure, a copper coil A and a wire are placed in the plane of the paper. A current of one ampere flows in the wire in the directions as shown in the figure. If the magnitude of the current is raised to 2 amp. What will be the direction of the induced current in the coil :-



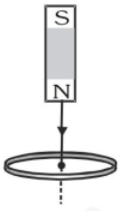
(A) anti clock wise

(B) clock wise

(C) no current will induced

(D) none of the above

Q9. Consider a metal ring kept on a horizontal plane. A bar magnet is held above the ring with its length along the axis of the ring. If the magnet is dropped freely the accelration of the falling magnet is (g is accelration due to gravity) :–



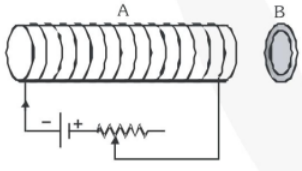
(A) more than g

(B) equal to g

(C) less than g

(D) depend on mass of magnet

Q10. An aluminium ring B faces an electromagnet A. The current I through A can be altered. Then which of the following statement is correct :–



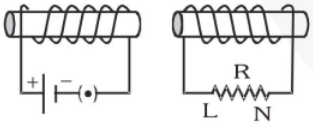
(A) if I decreases A will repel B

(B) whether I increases or decreases, B will not experience any force

(C) if I increases, A will repel B

(D) if I increases, A will attract B

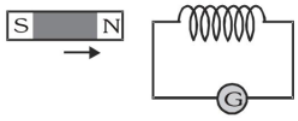
Q11. Two co–axial solenoids are shown in figure. If key of primary is suddenly opened then direction of instantaneous induced current in resistance 'R' in secondary is :-



(A) L to N (B) N to L

(C) alternating (D) zero

Q12. As shown in the figure, a magnet is brought towards a fixed coil quickly. Due to this the induced emf, current and the charge are E, I and Q respectively. If the speed of the magnet is doubled then which of the following statement is wrong :–



(A) E increases

(B) I increases

(C) Q does not change

(D) Q increases

Motional EMF

Q13. An aeroplane in which the distance between the tips of the wings is 50 meter is flying horizontally with a speed of 360 km/hour over a place where the vertical component of earth's magnetic field is 2.0×10^{-4} Testa. The potential difference between the tips of the wings would be :-

(A) 0.1 V (B) 1.0 V

(C) 0.2 V (D) 0.0

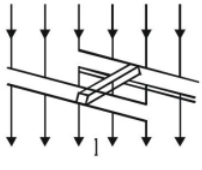
Q14. If the vertical component of earth's magnetic field be 6×10^{-5} Wb/m^2, then what will be the induced potential difference produced between the rails of a meter–gauge when a train is running on them with a speed of 36 km/h :–

(A) 2×10^{-4} V

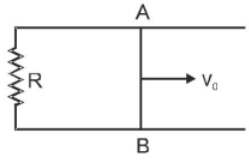
(B) 6×10^{-4} V

(C) 3×10^{-4} V

(D) 9×10^{-4} V



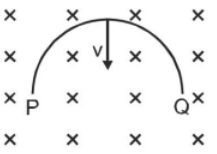
Q15. Two long parallel metallic wires with a resistance 'R' form a horizontal plane. A conducting rod AB is on the wires shown in figure. The space has magnetic field pointing vertically downwards. The rod is given an initial velocity 'v_0'. There is no friction in the wires and the rod. After a time 't' the velocity of the rod will be v such that :–



(A) $v > v_0$ (B) $v < v_0$

(C) $v = v_0$ (D) $v = -v_0$

Q16. A semicircle loop PQ of radius 'R' is moved with velocity 'v' in transverse magnetic field as shown in figure. The value of induced emf. at the end of loop is :-



(A) Bv (πr), end P at high potential

(B) 2BRv, end P at high potential

(C) 2BRv, end Q at high potential

(D) $B\frac{\pi R^2}{2}$ v, end P at high potential

Eddy Currents

Q17. Eddy current do not cause :-

(A) sparking

(B) loss of energy

(C) damping

(D) heating

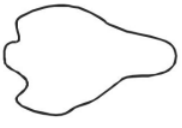
Q18. Plane of eddy circulations makes an angle with the external time varying field is :-

(A) 0° (B) 45°

(C) 180° (D) 90°

Induced Electric Field

Q19. As a result of change in the magnetic flux linked to the closed loop shown in the figure, an e.m.f. . V volt is induced in the loop. The work done (joules) in taking a charge Q coulomb once along the loop is :–



(A) QV (B) 0

(C) 2QV (D) $\frac{QV}{2}$

Self Inductance

Q20. When the current changes from 2A to 4A in 0.05 seconds in a coil, emf induced in the coil is 8V. The coefficient of self inductance in the coil will be :–

(A) 0.1 H (B) 0.2 H

(C) 0.4 H (D) 0.8 H

Q21 A coil of self inductance 2H carries a 2A current. If direction of current is reversed in 1 sec., then induced emf in it :-

(A) – 8V (B) 8V

(C) – 4V (D) Zero

Q22. Two coils are made of copper wires of same length In the first coil the number of turns is 3n and radius is r. In the second coil number of turns is n and radius is 3r the ratio of self inductances of the coils will be :-

(A) 9 : 1 (B) 1 : 1

(C) 1 : 3 (D) 1 : 9

Growth Of Current In LR Circuit

Q23. An LR circuit with a battery is connected at t = 0. Which of the following quantities is not zero just after the connection :-

(a) Current in circuit

(b) Magnetic potential energy in the inductor

(c) Power delivered by the battery

(d) Emf induced in the inductor

(A) a, b (B) a, c

(C) c, d (D) Only d

Q24. An L-R circuit consists of an inductance of 8 mH and a resistance of 4 Ω. The time constant of the circuit is :-

(A) 2 ms (B) 12 ms

(C) 32 ms (D) 500 s

Energy Stored In Inductor

Q25. A circular iron, core supports N turns. If a current I produces a magnetic flux ϕ across the core's cross section, then the magnetic energy is :-

(A) $I\phi$ (B) $\frac{1}{2} i\phi$

(C) $I\phi$ (D) $I^2\phi$

Q26. An inductor coil stores U energy when i current is passed through it and dissipates energy at the rate of P. The time constant of the circuit, when this coil is connected across a battery of zero internal resistance is -

(A) $\frac{4U}{P}$ (B) $\frac{U}{P}$

(C) $\frac{2U}{P}$ (D) $\frac{2P}{U}$

Decay Of Current In LR Circuit

Q27. **Statement–1** : At any instant, if the current through an inductor is zero, then the induced emf must be zero.

Statement–2 : An inductor tends to keep the flux constant.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Mutual Inductance

Q28. Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon-

(A) the rates at which currents are changing in the two coils

(B) relative position and orientation of the two coils

(C) the materials of the wires of the coils

(D) the currents in the two coils

Q29. The mutual inductance of the two coils is 1.25 henry. In the first coil, the current changes at 80 amp/s, the value of induced emf in volts in the secondary coil will be :-

(A) 12.5 (B) 64

(C) 0.016 (D) 100

Inductors In Series And Parallel

Q30. The equivalent inductance of two inductances is 2.4 henry when connected in parallel and 10 henry when connected in series. The difference between the two inductances is:-

(A) 2 henry (B) 3 henry

(C) 4 henry (D) 5 henry

AC Generator

Q31. In an AC generator, a coil with N turns, all of the same area A and total resistance R, rotates with frequency ω in a magnetic field B. The maximum value of emf generated in the coil is-

(A) NABRω (B) NAB

(C) NABR (D) NABω

Q32. The working of dynamo is based on principle of :-

(A) electromagnetic induction

(B) conversion of mechanical energy into electricity

(C) magnetic effects of current

(D) heating effects of current

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The e.m.f. induced in a coil of wire, which is rotating in a magnetic field, does not depend on

(A) the angular speed of rotation

(B) the area of the coil

(C) the number of turns on the coil

(D) the resistance of the coil

Q2. A conducting loop is placed in a uniform magnetic field with its plane perpendicular to the field. An emf is induced in the loop if :-

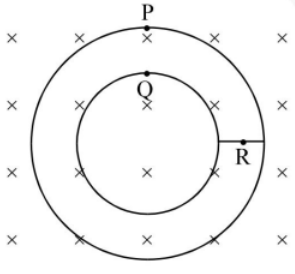
(A) It is translated within magnetic field

(B) It is rotated about its axis

(C) It is rotated about a diameter

(D) It is deformed

Q3. Figure shown plane figure made of a conductor located in a magnetic field along the inward normal to the plane of the figure. The magnetic field starts diminishing. Then the induced current



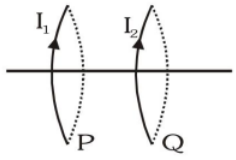
(A) at point P is clockwise

(B) at point Q is anticlockwise

(C) at point Q is clockwise

(D) at point R is zero

Q4. Two circular coils P & Q are fixed coaxially & carry currents I_1 and I_2 respectively.



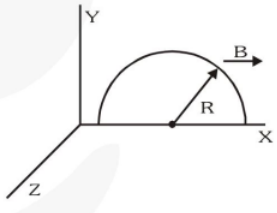
(A) If $I_2 = 0$ & P moves towards Q, a current in the same direction as I_1 is induced in Q.

(B) If $I_1 = 0$ & Q moves towards P, a current in the opp. direction to that of I_2 is induced in P

(C) when $I_1 \neq 0$ and $I_2 \neq 0$ are in the same direction then the two coils tend to move apart.

(D) when $I_1 \neq 0$ and $I_2 \neq 0$ are in opposite directions then the coils tends to move apart

Q5. A semicircle conducting ring of radius R is placed in the xy plane, as shown in the figure. A uniform magnetic field is set up along the x–axis. No emf, will be induced in the ring. If



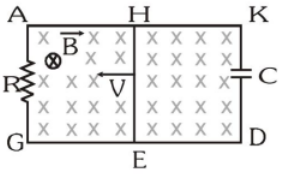
(A) it moves along the x–axis

(B) it moves along the y–axis

(C) it moves along the z–axis

(D) All of these

Q6. In the circuit shown in figure, a conducting wire HE is moved with a constant speed V towards left. The complete circuit is placed in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the circuit directed in inward direction.



The current in HKDE is -

(A) clockwise

(B) anticlockwise

(C) alternating

(D) zero

Q7. When a metallic sphere is moved in a magnetic field, it gets heated due to :–

(A) direct current

(B) eddy currents

(C) alternating current

(D) additional current

Q8. A conducting ring of radius 1 meter is placed in an uniform magnetic field B of 0·01 Tesla oscillating with frequency 100 Hz with its plane at right angles to B. What will be the induced electric field:

(A) π volts/m (B) 2π volts/m

(C) 10 volts/m (D) 62 volts/m

Q9. Self induction of solenoid is :-

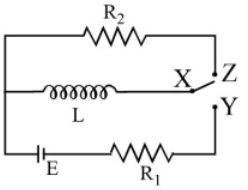
(A) directly proportional to current flowing through the coil

(B) directly proportional to its length

(C) directly proportional to area of cross–section

(D) inversely proportional to area of cross-section

Q10. In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :



(A) $\frac{LE^2}{2R_1^2}$ (B) $\frac{LE^2}{2R_2^2}$

(C) $\frac{LE^2}{2R_1R_2}$ (D) $\frac{LE^2R_2}{2R_1^2}$

Q11. Two coils have a mutual inductance 0.005H. The current changes in the first coil according to the equation $I = I_0 \sin\omega t$, where $I_0 = 10A$ and $\omega = 100\pi$ rad/s. The maximum value of emf in the second coil will be :–

(A) 5π (B) 10π (C) 2.5π (D) 15π

Q12. Magnetic field which keeps the particles in circular path of fixed radius keeping speed constant must :-

(A) Remain a constant every where

(B) Increase gradually

(C) Decrease gradually

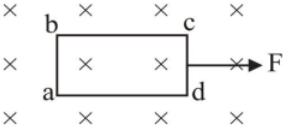
(D) None of these

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for question nos. 13 to 15

A rectangular frame of wire abcd has dimensions 32 cm × 8.0 cm and a total resistance of 2.0 Ω. It is pulled out of a magnetic field B = 0.020 T by applying a force of 3.2×10^{-5} N (figure). It is found that the frame moves with constant speed.



Q13. The constant speed of the frame is–

(A) 25 m/s (B) 20 m/s

(C) 16 m/s (D) 7.5 m/s

Q14. The emf induced in the loop is–

(A) 8.0×10^{-2} V (B) 5.0×10^{-2} V

(C) 2.0×10^{-2} V (D) 4.0×10^{-2} V

Q15. The potential difference between the points a and b is equal to -

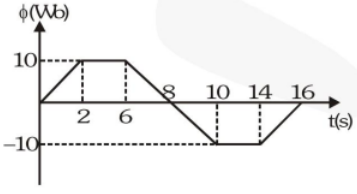
(A) 4.8×10^{-2} V (B) 3.6×10^{-2} V

(C) 2.4×10^{-2} V (D) 1.6×10^{-2} V

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q16. Magnetic flux in a circular coil of resistance 10Ω changes with time as shown in figure. ⊗ direction indicates a direction perpendicular to paper inwards. Match the following table :





Column–I	Column–II
(A) At 1s induced current is	(p) clockwise
(B) At 5s induced current is	(q) anticlockwise
(C) At 9s induced current is	(r) zero
(D) At 15 s induced current is	(s) 2A
	(t) None of these

NUMERICAL TYPE

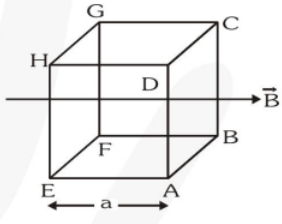
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q17. The magnetic field perpendicular to the plane of a loop of area 0.1 m^2 is 0.2 T. Calculate the magnetic flux through the loop (in wb).

SUBJECTIVE TYPE

Q18. A coil of 100 turns and area 5cm^2 is placed in a external magnetic field of 0.2 Tesla (S.I), in such a way that it makes an angle 30° with the field direction. Calculate magnetic flux through the coil (in weber).

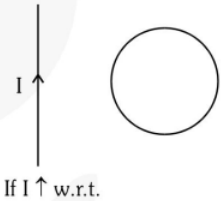
Q19. A square cube of side 'a' is placed in uniform magnetic field 'B'. Find magnetic flux through each face of the cube.



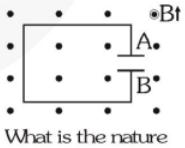
Q20. Find direction of induced current for the given cases :–

(Where w.r.t. = with respect to time, ob = observer =)

(i)

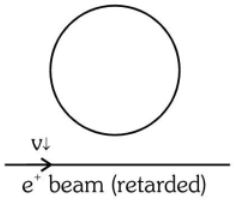


(ii)

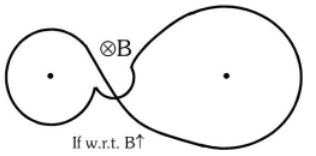


What is the nature of charge on plates of capacitor ?

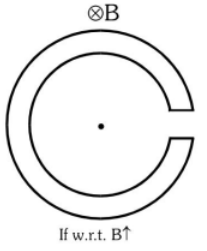
(iii)



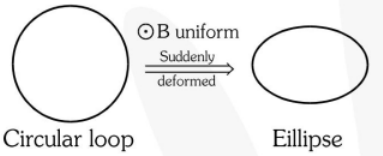
(iv) (a)



(b)

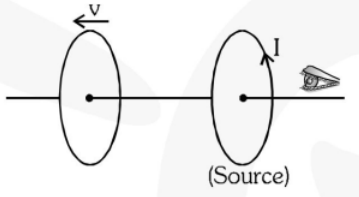


(v)

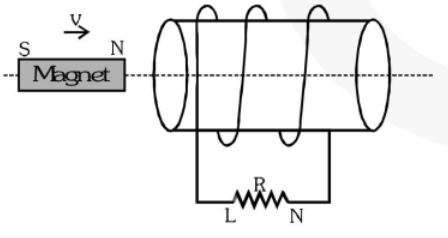


Q21. Find direction of induced current in the given cases :–

(i) (a)



(ii) What is the direction of induced current in resistance 'R' ?



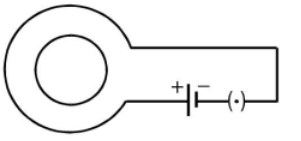
Q22. Find direction of instantaneous induced current in secondary circuit for the following changes in primary circuit :-

(a) Key is just closed

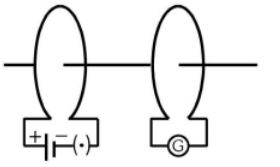
(b) Some time after the closing of key

(c) Key is just opened

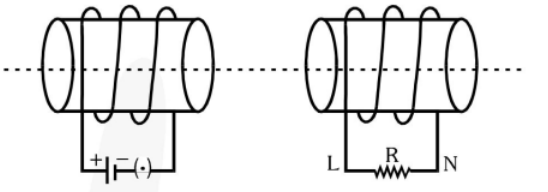
(i)



(ii)

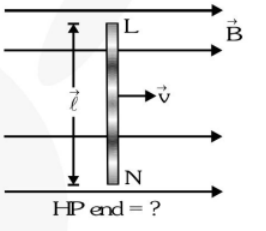


(iii)



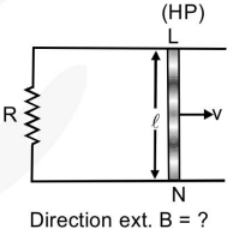
Q23. Find the Given Parameter when straight conductor moves in external magnetic field:-

(i)

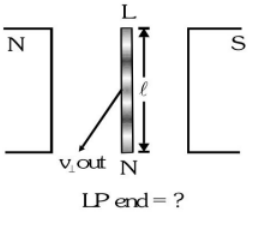


High Potential End

(ii)



(iii)



low Potential End

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A ring of resistance 10Ω, radius 10cm and 100 turns is rotated at a rate 100 revolutions per second about its diameter is perpendicular to a uniform magnetic field of induction 10mT. The amplitude of the current in the loop will be nearly (Take : $\pi^2 = 10$)

(A) 200A (B) 2A

(C) 0.002A (D) none of these

Q2. A coil and a magnet moves with their constant speeds 5 m/sec. and 3 m/sec. respectively, towards each other, then induced emf in coil is 16 mV. If both are moves in same direction, then induced emf in coil:-

(A) 15 mV (B) 4 mV

(C) 64 mV (D) Zero

Q3. The radius of a circular coil having 50 turns is 2 cm. Its plane is normal to the magnetic field. The magnetic field changes from 2T to 4T in 3.14 sec. The induced emf in coil will be :–

(A) 0.4V (B) 0.04V

(C) 4 mV (D) 0.12 V

Q4. A short bar magnet allowed to fall along the axis of horizontal metallic ring. Starting from rest, the distance fallen by the magnet in one second may be :–

(A) 4.0 m. (B) 5.0 m.

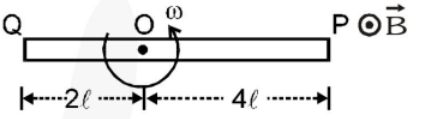
(C) 6.0 m. (D) 7.0 m.

Q5. Two parallel long straight conductors lie on a smooth surface. Two other parallel conductors rest on them at right angles so as to form a square of side a initially. A uniform magnetic field B exists at right angles to the plane containing the conductors. They all start moving out with a constant velocity v. If r is the resistance per unit length of the wire the current in the circuit will be

(A) $\frac{Bv}{r}$ (B) $\frac{Br}{v}$

(C) Bvr (D) Bv

Q6. A conducting rod rotates with a constant angular velocity 'ω' about the axis which passes through point 'O' and perpendicular to its length. A uniform magnetic field 'B' exists parallel to the axis of the rotation. Then potential difference between the two ends of the rod is :-



(A) $6B\omega\ell^2$ (B) $B\omega\ell^2$

(C) $10B\omega\ell^2$ (D) zero

Q7. A semicircular wire of radius R is rotated with constant angular velocity ω about an axis passing through one end and perpendicular to the plane of the wire. There is a uniform magnetic field of strength B. The induced e.m.f. between the ends is -



(A) $B\omega R^2/2$ (B) $2B\omega R^2$

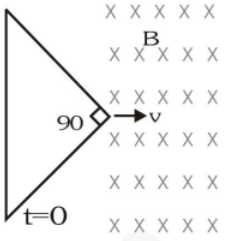
(C) is variable (D) none of these

Q8. The magnetic field in a region is given by $\vec{B} = B_0\left(1+\frac{x}{a}\right)\hat{k}$. A square loop of edge length d is placed with its edge along x & y axis. The loop is moved with constant velocity $\vec{v} = V_0\hat{i}$. The emf induced in the loop is

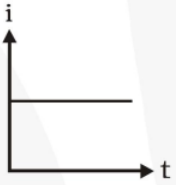
(A) $\frac{V_0B_0d^2}{a}$ (B) $\frac{V_0B_0d^2}{2a}$

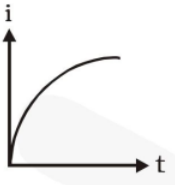
(C) $\frac{V_0B_0a^2}{d}$ (D) none

Q9. The figure shows an isosceles triangle wire frame with apex angle equal to $\pi/2$. The frame starts entering into the region of uniform magnetic field B with constant velocity v at t = 0. The longest side of the frame is perpendicular to the direction of velocity. If i is the instantaneous current through the frame then choose the alternative showing the correct variation of i with time.



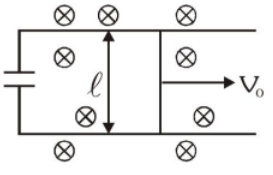
(A) 

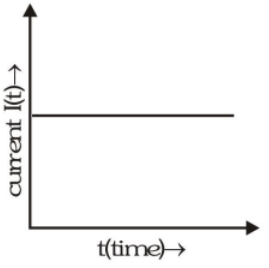
(B) 

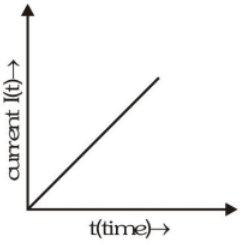
(C) 

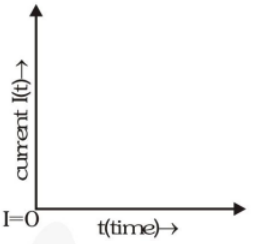
(D) 

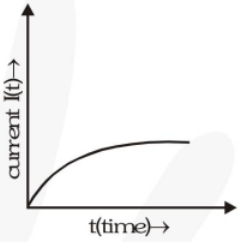
Q10. Two infinitely long conducting parallel rails are connected through a capacitor C as shown in the figure. A conductor of length ℓ is moved with constant speed v_0 in uniform magnetic field B. Which of the following graph truly depicts the variation of current through the conductor with time?



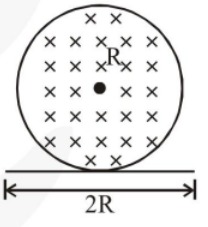
(A) 

(B) 

(C) 

(D) 

Q11. A uniform but time varying magnetic field is present in a circular region of radius R. The magnetic field is perpendicular and into the plane of the loop and the magnitude of field is increasing at a constant rate α. There is a straight conducting rod of length 2R placed as shown in figure. The magnitude of induced emf across the rod is



(A) $\pi R^2\alpha$ (B) $\frac{\pi R^2\alpha}{2}$

(C) $\frac{R^2\alpha}{\sqrt{2}}$ (D) $\frac{\pi R^2\alpha}{4}$

Q12. The self inductance of a toroid is :-

(A) $\frac{\mu_0 N^2 r^2}{2\,R_m}$ (B) $\frac{\mu_0 N^2 \pi r}{2R_m}$

(C) $\frac{\mu_0 N^2 r}{2R_m}$ (D) $\frac{\mu_0 N^2 r\pi}{R_m}$

Q13. A wire of fixed length is wound on a solenoid of length 'ℓ' and radius 'r'. Its self inductance is found to be L. Now if same wire is wound on a solenoid of length $\ell/2$ and radius r/2, then the self inductance will be -

(A) 2 L (B) L (C) 4 L (D) 8 L

Q14. An e.m.f. of 15 volt is applied in a circuit containing 5 henry inductance and 10 ohm resistance. The ratio of the currents at time $t \to \infty$ and at t = 1 second is:-

(A) $\frac{e^{1/2}}{e^{1/2}-1}$ (B) $\frac{e^{2}}{e^{2}-1}$

(C) $1-e^{-1}$ (D) $e^{-1}d$

Q15. A coil of $L=5\times10^{-3}$ H and R=18 Ω is abruptly supplied a potential of 5 volts. What will be the rate of change of current in 0.001 second ? ($e^{-3.6} = 0.0273$)

(A) 27.3 amp/sec. (B) 27.8 amp/sec.

(C) 2.73 amp/sec. (D) 2.78 amp/sec.

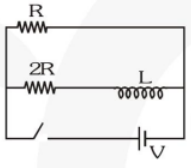
Q16. The ratio of time constant in charging and discharging in the circuit shown in figure is

(A) 1:1

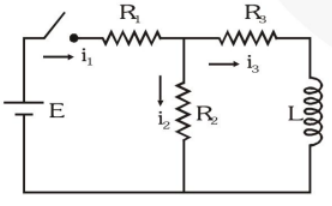
(B) 3:2

(C) 2:3

(D) 1:3



Q17. In the circuit shown in adjoining fig E = 10V, $R_1 = 1\Omega$ $R_2 = 2\Omega$, $R_3 = 3\Omega$ and L = 2H. Calculate the value of current i_1, i_2 and i_3 immidiately after key S is closed :-



(A) 3.3 amp, 3.3 amp, 3.3 amp

(B) 3.3 amp, 3.3 amp, 0 amp

(C) 3.3 amp, 0 amp, 0 amp

(D) 3.3 amp, 3.3 amp 1.1 amp

Q18. If a current of 2A give rise a magnetic flux of 5×10^{-5} weber/turn through a coil having 100 turns, then the magnetic energy stored in the medium surrounding by the coil is :-

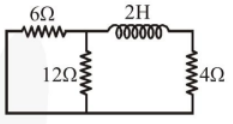
(A) 5 joule

(B) 5×10^{-7} joule

(C) 5×10^{-3} joule

(D) 0.5 joule

Q19. In the given circuit containing an inductor and resistors the time constant of the circuit is :-



(A) $\frac{1}{2}$s (B) $\frac{1}{8}$s

(C) $\frac{1}{6}$s (D) $\frac{1}{4}$s

Q20. Two coils of self inductances 2 mH and 8 mH are placed so close together that the effective flux in one coil is completely linked with the other. The mutual inductance between these coils is :-

(A) 10 mH (B) 6 mH

(C) 4 mH (D) 16 mH

Q21. An inductance coil have the time constant 4 sec. If it is cut into two equal parts and connected in parallel then new time constant of the circuit is :-

(A) 4 sec (B) 2 sec

(C) 1 sec (D) 0.5 sec

Q22. A circular coil of mean radius of 7 cm and having 4000 turns is rotated at the rate of 1800 revolutions per minute in the earth's magnetic field (B = 0.5 gauss), the peak value of emf induced in coil will be : (Take $\pi^2 = 10$)

(A) 1.158 V (B) 0.588 V

(C) 0.29 V (D) 5.8 V

ONE OR MORE THAN ONE CORRECT TYPE

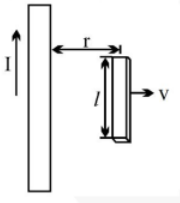
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The radius of a coil decreases steadily at the rate of 10^{-2} m/s. A constant and uniform magnetic field of induction 10^{-3} Wb/m^2 acts perpendicular to the plane of the coil. The radius of the coil when the induced e.m.f. in the coil is 1μV, is :-

(A) $\frac{2}{\pi}$ cm (B) $\frac{3}{\pi}$ cm

(C) $\frac{4}{\pi}$ cm (D) $\frac{5}{\pi}$ cm

Q2. A conducting rod moves with constant velocity υ perpendicular to the long, straight wire carrying a current I as shown. Compute the emf generated between the ends of the rod.



(A) $\frac{\mu_0 v I l}{\pi r}$

(B) $\frac{\mu_0 v I l}{2\pi r}$

(C) $\frac{2\mu_0 v I l}{\pi r}$

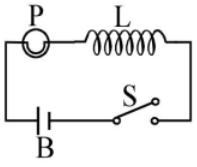
(D) $\frac{\mu_0 v I l}{4\pi r}$

Q3. A coil of 10^{-2} H inductance carries a current I = 2sin(100t)A. When current is half of its peak value then at that instant the induced emf in coil :-

(A) 1V (B) $\sqrt{2}$ V

(C) $\sqrt{3}$ V (D) 2V

Q4. In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to -



(A) the low resistance of P

(B) the induced-emf in L

(C) the low resistance of L

(D) the high voltage of the battery B

Q5. A long straight wire of circular cross-section is made of a non-magnetic material. The wire is of radius a. The wire carries a current I which is uniformly distributed over its cross-section. The energy stored per unit length in the magnetic field contained within the wire is.

(A) $U = \frac{\mu_0 I^2}{8\pi}$ (B) $U = \frac{\mu_0 I^2}{16\pi}$

(C) $U = \frac{\mu_0 I^2}{4\pi}$ (D) $U = \frac{\mu_0 I^2}{2\pi}$

Q6. Two coil A and B have coefficient of mutual inductance M=2H. The magnetic flux passing through coil A changes by 4 Weber in 10 seconds due to the change in current in B. Then -

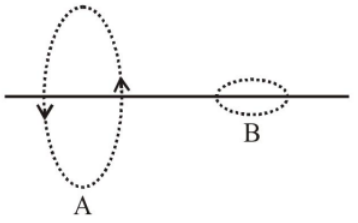
(A) Change in current in B in this time interval is 0.5 A.

(B) The change in current in B in this time interval is 2A.

(C) The change in current in B in this time interval is 8A.

(D) A change in current of 1A in coil A will produce a change in flux passing through B by 4 Weber.

Q7. In the given figure the centre of a small conducting circular loop B lies on the axis of bigger circular loop A and their axis are mutually perpendicular. An anticlockwise (when viewed from the side of B) current in the loop A start increasing then :-



(A) current induced in the loop B is in clockwise direction (when viewed from above the B)

(B) current induced in the loop B is in anti-clockwise direction (when viewed from above the B)

(C) current must induced in the loop B but its direction can not be predicted

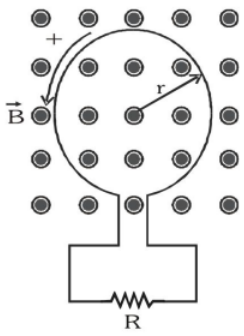
(D) no current is induced in the loop B

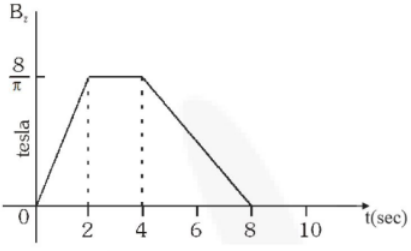
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

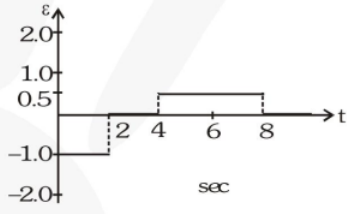
Paragraph for question nos. 8 to 9

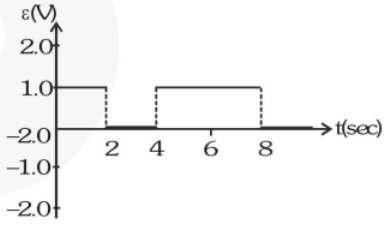
A uniform magnetic field $\vec{B}$ is perpendicular to a one-turn circular loop of wire of negligible resistance, as shown in the figure given below. The field changes with time as shown (the z direction is out of the page). The loop is of radius r = 50 cm and is connected in series with a resistor of resistance R = 20Ω. The "+" direction around the circuit is indicated in the figure.

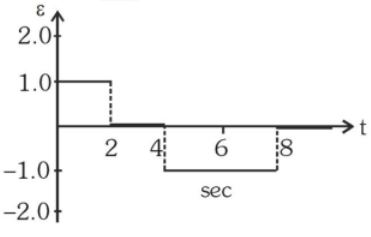


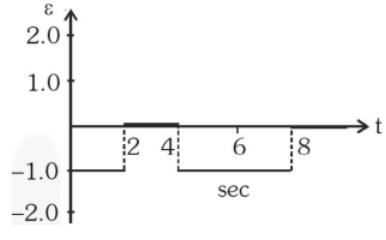


Q8. Graph of induced emf as a function of time is given by :-

(A)


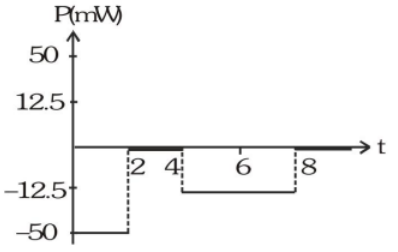
(B)


(C)


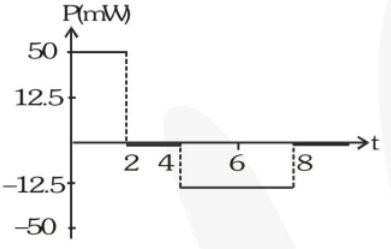
(D)


Q9. Plot the power dissipated in the resistor as a function of time.

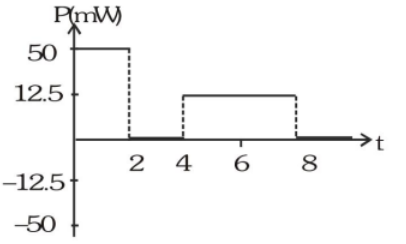
(A)



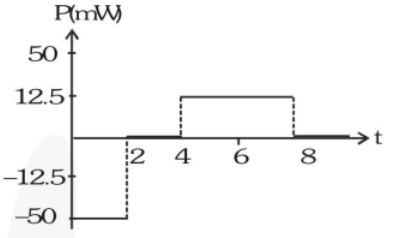
(B)



(C)



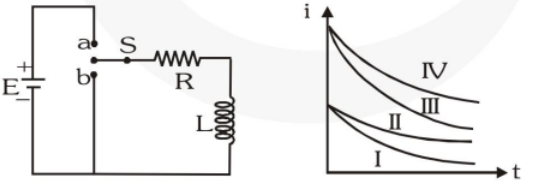
(D)



MATCH THE COLUMN TYPE

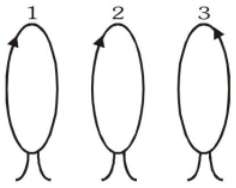
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10. The switch S in the circuit is connected with point 'a' for a very long time, then it is shifted to position 'b'. The resulting current through the inductor is shown by curves in the graph for four sets of values for the resistance R and inductance L (given in the left column). Which set corresponds with which curve?



Column I	**Column II**
(A) R_0 and L_0	(p) I
(B) $2R_0$ and L_0	(q) II
(C) R_0 and $2L_0$	(r) III
(D) $2R_0$ and $2L_0$	(s) IV

Q11. Three coils are placed infront of each other as shown. Currents in 1 and 2 are in same direction, while that in 3 is in opposite direction. Match the following table :



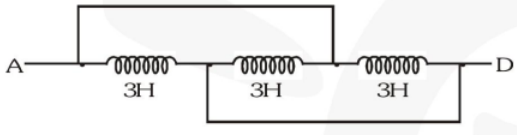
Column-I	**Column-II**
(a) When current in 1 is increased	(p) current in 1 will increase
(b) When current in 2 is increased	(q) current in 2 will increase
(c) When current in 3 is increased	(r) Current in 3 will increase
	(s) none of these

(A) (a - q), (b - p, r), (c -s)
(B) (a - r), (b - r), (c -p, q)
(C) (a - r), (b - q), (c -p)
(D) (a - s), (b - r), (c q)

NUMERICAL TYPE

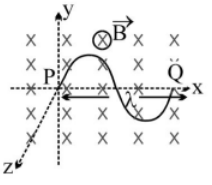
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q12. The inductance between A and D is-



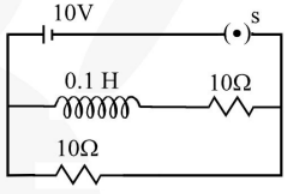
SUBJECTIVE TYPE

Q13. A wire forming one cycle of sine curve is moved in x-y plane with velocity $\vec{V} = V_x\hat{i} + V_y\hat{j}$. There exist a magnetic field $\vec{B} = -B_0\hat{k}$. Find the motional emf develop across the ends PQ of wire.



Q14. In a L–R decay circuit, the initial current at t = 0 is I. The total charge that has flown through the resistor till the energy in the inductor has reduced to one–fourth its initial value, is-

Q15. In the adjoining circuit, initially the switch S is open. The switch 'S' is closed at t = 0. The difference between the maximum and minimum current that can flow in the circuit is



Q16. An inductor of inductance 2.0mH,is connected across a charged capacitor of capacitance 5.0μF and the resulting LC circuit is set oscillating at its natural frequency. Let Q denote the instantaneous charge on the capacitor and I be the current in the circuit .It is found that the maximum value of Q is 200μC.

(a) when Q = 100μC, what is the value of $|dI/dt|$?

(b) when Q = 200 μC, what is the value of I ?

(c) Find the maximum value of I.

(d) when I is equal to one half its maximum value, what is the value of $|Q|$

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	B	B	B, D	A	B	B	B	C	C
11	12	13	14	15	16	17	18	19	20
A	A	B	B	B	C	A	D	A	B
21	22	23	24	25	26	27	28	29	30
B	B	D	A	B	C	D	B	D	A
31	32								
D	A								

Ex-1A

1. D **2.** C,D **3.** A,C,D **4.** B, D **5.** D

6. D **7.** B **8.** A **9.** C **10.** A

11. A **12.** A **13.** A **14.** D **15.** B

16. (A) → (q) ; (B) → (r) ; (C) → (p) ; (D) → (q)

17. 0.02 Wb **18.** 5×10^{-3} weber. **19.** $\boxed{\phi_{net} = 0}$

20. (i) Anticlockwise
(ii) A – Positive charge, B – Negative charge
(iii) Anticlockwise
(iv) (a) Anti clockwise in bigger loop & clockwise in smaller loop
(b) Anti clockwise in bigger loop & clockwise in smaller loop
(v) Anticlockwise

21. (i) (a) Anticlockwise (ACW),
(b) Clockwise (CW)
(ii) N to L

22. (i) (a) ACW, (b) Zero, (c) CW
(ii) (a) ACW, (b) Zero, (c) CW
(iii) (a) L to N, (b) Zero, (c) N to L

23. (i) $e = 0$ as $\vec{v} \times \vec{B} = 0$
(ii) Magnetic field will be downward perpendicular to the plane paper ($\otimes$).
(iii) N will be at LP.

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	B	B	A	A	A	B	A	D	C
11	12	13	14	15	16	17	18	19	20
D	A	A	B	A	B	B	C	D	C
21	22								
A	B								

Ex-2A

1. D **2.** B **3.** C **4.** B **5.** B

6. B **7.** D **8.** A **9.** C

10. (A) → (r); (B) → (p); (C) → (s); (D) → (q) **11.** B **12.** 1H

13. $\lambda V_y B_0$ **14.** $LI/2R$ **15.** 1A

16. (a) 10^4A/s (b) 0 (c) 2A (d) $100\sqrt{3}\,\mu$C

-: Important Notes & Formulas :-